

Akhil Arya

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Education

University of Wisconsin–Madison <i>Master of Science - MS, Computer Science</i>	2026-Present
Indian Institute of Technology (IIT) Ropar <i>B.Tech, Computer Science and Engineering</i>	2019-2023

Experience

Microsoft <i>Software Engineer</i> Worked on systems that are an integral part of Microsoft Commerce ecosystem and are responsible for subscription management of volume licensing customers of non-Azure Microsoft products like Office 356, Dynamics 365, Intune, MS Teams etc. Led key initiatives such as: • Simplified multi-tenant provisioning by integrating legacy commerce systems with modern commerce APIs to seamlessly transfer agreement billing data, enabling unified provisioning and reducing manual overhead. • Refactored and optimized request handling service to reduce SQL server bottlenecks during quarter-close load spikes, reducing query latency and message queue backlog across downstream services by 90%. • Enhanced system security and reliability by collaborating across teams to expand error observability through detailed event and request-flow logging and implementing proactive monitoring frameworks with automated alerting, improving failure detection time by 99.97%. • Drove key security initiatives, including threat modeling, privacy compliance, and migrating authentication from certificate-based mechanisms to Azure Active Directory (AAD). • Performed Code Integrity Enforcement across 50+ Azure VMs hosting production services, ensuring that only Microsoft-signed and verified binaries are executed in runtime environments. Integrated code-signing into build pipelines, removed unwanted artifacts, and updated legacy NuGet packages. • Built and deployed an SRE agent that automated incident triage and resolution, reducing DRI effort by 50%, and developed an AI agent to convert quarterly org requirements into auto-published sprint deliverables. • Migrated classic release pipelines to EV2 pipelines, which offer upgraded security features. • Resolved 100+ high-impact customer escalations through deep debugging and root-cause analysis. • Technologies: C#, .NET Core, Azure, Azure DevOps, SQL Server, EV2, Kusto (KQL), GHCP, MCP, NuGet.	Jun 2023 – Jan 2026 <i>Noida, India</i>
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Projects

Nexus • Built Nexus, an autonomous multi-agent systems engineering kernel capable of executing long-horizon (1000+ step) workflows by combining Graph RAG memory, Model Context Protocol (MCP) toolchains, and agentic planning without relying on paid APIs. • Engineered self-extensible agent infrastructure with dynamic MCP tool generation, hierarchical memory graphs (Neo4j + Qdrant), and MCTS-based decision-making, enabling runtime capability expansion, efficient context retention, and autonomous task execution across complex environments. • Technologies: Python, LLMs, AI Agents, Model Context Protocol (MCP), Graph RAG, Neo4j, Qdrant, OpenTelemetry, Jaeger, JSON-RPC	May 2026 - June 2026
Sampan App and Dashboard • Created an app and dashboard for the Ropar District Administration to record and visualize child healthcare data, processing 100K+ entries per month across all blocks. • Technologies and Frameworks: Flutter, Firebase, Google Apps Script, Bootstrap, Flask and Node.js.	Jan 2022 - May 2022

Technical Skills

Libraries/ Framework: .NET Core, MySQL, PostgreSQL, KQL, Azure, NumPy, OpenCV, Matplotlib, Node.js, React.js
Programming Languages: C++, C#, Python, HTML, CSS, JavaScript, Java
Others: Data Structures, Algorithms, System Design, Operating System, DBMS, OOPS, Artificial Intelligence, Machine Learning, Git, Copilot, LLM, MCP